SIMULATED
Asset Management

Portfolio insights and benchmarking

Agent: **Portfolio Management**

OPENING PROMPT

```
What are my top 10 holdings by market value in the SAM Technology & Infrastructure
portfolio?
```

The demo date is the latest price date in the Snowflake Public Data share at build time, and positions are month-end. Agents answer from live data, so figures, names and rankings in your environment can differ from the examples in this script.

Business Context Setup

Persona: Anna, Senior Portfolio Manager at Simulated Asset Management
Business Challenge: Portfolio managers need instant access to portfolio analytics, holdings information, and supporting research to make informed investment decisions. Traditional systems require multiple tools, manual data gathering, and time-consuming analysis that delays critical investment decisions.
Value Proposition: AI-powered portfolio analytics that combines quantitative holdings data with qualitative research insights in seconds, enabling faster decision-making and better risk management.

Agent: `AM_portfolio_management_copilot`
Data Available: 11 portfolios, month-end holdings, broker research, earnings transcripts and press releases, firm policy documents, live compliance alerts, transaction-cost and liquidity data

Demo Flow

Scene Setting: Anna is preparing for her weekly portfolio review meeting and needs to quickly assess her Technology & Infrastructure portfolio, understand current holdings, and identify any concentration risks that require attention.

Presenter Note: The five prompts are one conversation - run them in the same thread so each follow-up can lean on the previous answer. The agent often answers holdings questions by running semantic-view SQL directly rather than through a visibly named `portfolio_analyzer` call; the data source is the same `SAM_PORTFOLIO_VIEW`.

Step 1: Top Holdings Overview

User Input:

PROMPT

```
What are my top 10 holdings by market value in the SAM Technology & Infrastructure
portfolio?
```

Tools Used:

- `portfolio_analyzer` (Cortex Analyst) / SQL - Query holdings from SAM_PORTFOLIO_VIEW at the latest month-end
- `data_to_chart` - Bar chart of the top 10

Expected Response:

- Table and bar chart of the top 10 with ticker, name, sector, market value and weight, as at the latest month-end
- **Apple, Microsoft and NVIDIA are the three largest positions**, all above the 6.5% concentration warning level; Alphabet, AMD, Broadcom, Costco, Amazon, TSMC and Commercial Metals make up the rest of the top 10
- Top 3 are roughly a quarter of the portfolio and the top 10 close to 60%

- The agent may point out the non-tech names (Costco, Commercial Metals) as the "& Infrastructure" tilt, and closes by offering a sector breakdown or concentration check

Talking Points:

- Instant portfolio analytics without SQL or complex queries
- The concentration story is visible from the first answer
- Positions served straight from the warehouse at the latest month-end, with no extract or reconciliation step

Key Features Highlighted:

- Cortex Analyst semantic understanding of portfolio data
- Natural language to SQL conversion
- Automatic charting of tabular results

Step 2: Latest Research for Top Holdings

Presenter Transition:

> *"We can see our top holdings clearly - Apple, Microsoft and NVIDIA are the three largest and all sit above the concentration watch threshold. But portfolio data alone doesn't tell the full story - what are the analysts actually saying about these large positions? Let me show you how we seamlessly transition from quantitative data to qualitative research..."*

Reasoning: Quantitative holdings data raises questions that qualitative research can answer. This demonstrates the multi-modal capability of combining analytics with document search.

User Input:

> **PROMPT**
>
> ```
> Based on those top holdings you just showed me, what is the latest broker research saying about our three largest positions?
> ```

Tools Used:

- `search_external_docs` (Cortex Search) - Broker research filtered by ticker for the three names from Step 1 (one call, or one per name)

Expected Response:

- Latest broker note for each of Apple, Microsoft and NVIDIA, with fictional broker name, date, rating and 12-month price target
- **Microsoft is the only Buy; Apple and NVIDIA are rated Hold**, held back by valuation rather than fundamentals
- Price targets have been raised steadily through the year for all three, and the agent shows the target trail
- Short thesis summary per name (growth, margins, key risks such as competitive intensity and enterprise tech-spend sensitivity)

Talking Points:

- The agent picks up "our three largest positions" from Step 1 without being told the names
- Seamless transition from quantitative holdings data to qualitative research insights
- The nuance matters: the two largest positions carry the least conviction from the brokers

Key Features Highlighted:

- Contextual follow-up that builds on previous query results
- Ticker-filtered search linking holdings to research
- Automatic citation and source attribution

Step 3: Sector Risk Assessment

Presenter Transition:

> *"Only Microsoft gets a Buy - Apple and NVIDIA are Holds. And having three large technology positions creates a different kind of risk: sector concentration. Let me show you how we can assess sector-level risk while keeping the context of our individual holdings..."*

Reasoning: Individual position analysis (Step 1) and research sentiment (Step 2) need to be viewed through a sector concentration lens to understand cumulative risk exposure.

User Input:

> **PROMPT**
>
> ```
> Looking at those top holdings and their research, what's our sector concentration risk in this portfolio, especially for the companies with the largest positions?
> ```

Tools Used:

- `concentration-risk-assessment` skill
- `portfolio_analyzer` (Cortex Analyst) / SQL - Sector allocation from SAM_PORTFOLIO_VIEW
- `search_internal_docs` (Cortex Search) - Concentration Risk Management Policy and IPS limits
- `data_to_chart` - Sector allocation chart

Expected Response:

- Sector allocation with **Information Technology at about half the portfolio**, far above the next sectors
- Limits read from firm policy: **6.5% single-position warning, 7.0% breach** (Concentration Risk Management Policy), plus IPS single-issuer and single-sector guidelines
- **Apple, Microsoft and NVIDIA flagged as above the 7.0% breach line**; AMD and Broadcom noted against the single-issuer guideline
- The compounding risk: all three flagged names are IT, so a tech drawdown hits single-name and sector risk at once
- The agent usually caveats that this is a tech-thematic mandate, so a sector tilt is partly by design

Talking Points:

- Integrated risk analysis that combines individual position risk (Step 1) with sector risk
- Research sentiment (Step 2) now viewed through a concentration lens
- Thresholds come from the firm's own policy documents, not from the prompt

Key Features Highlighted:

- Multi-dimensional risk analysis building on previous queries
- Policy retrieval combined with live holdings
- Skill-guided concentration workflow

Step 4: Integrated Risk & Action Plan

Presenter Transition:

> *"Now we have the complete risk picture - a breach on three names, all in a sector that is half the book, with mixed research support. The real value comes from synthesising all of this into actionable recommendations. Let me ask the agent to prioritise which positions need attention..."*

Reasoning: Analysis without action is incomplete. This step demonstrates how the AI synthesises multi-step analysis into prioritised, actionable recommendations.

User Input:

> **PROMPT**
>
> ```
> Based on our concentration analysis and research findings, which of our largest positions need attention and what actions should we consider?
> ```

Tools Used:

- `portfolio_analyzer` (Cortex Analyst) / SQL - Live compliance alerts for the portfolio from SAM_PORTFOLIO_VIEW
- Policy thresholds and research from Steps 2-3 are reused from the thread rather than searched again

Expected Response:

- The agent checks the live compliance alerts and finds **two active concentration breaches: Apple and Microsoft**, raised against the mandate's own 8.0% single-issuer limit (an aggressive IPS profile, looser than the firm-wide 7.0%)
- Prioritised list:
  - **Apple - trim first**: largest position, widest breach, and only a Hold from its broker
  - **Microsoft - minimal trim to clear the breach**: in breach, but the only Buy, so the research argues for keeping as much exposure as possible
  - **NVIDIA - watch, do not add**: no active alert, but in the warning zone and rated Hold
  - **Sector overlay**: trimming the single names barely dents the IT weight; recycle proceeds outside IT if the committee wants to cut the systematic bet
- Closes by offering a counterfactual or a full implementation plan

Talking Points:

- The agent reconciles its own earlier caveat: firm-wide thresholds versus the mandate-specific limit recorded on the live alerts
- Research conviction changes the action: same breach, different trim size for Apple and Microsoft
- Professional portfolio management workflow from analysis to action

Key Features Highlighted:

- Comprehensive decision support building on multi-step analysis
- Integration of position size, sector risk, research sentiment and live compliance state
- Actionable recommendations with clear prioritisation and rationale

Step 5: Portfolio Management Decision

Presenter Transition:

> *"We know what needs attention. But professional portfolio management requires more than identifying issues - we need specific trades with sizes, costs and timelines. Let me show you how the agent translates recommendations into an implementation plan..."*

Reasoning: Recommendations need to become executable actions. This final step demonstrates end-to-end workflow from analysis to specific implementation.

User Input:

PROMPT

```
Based on our complete analysis, provide me with a specific implementation plan including exact position sizes, timelines, and dollar amounts for the portfolio actions we should take.
```

Tools Used:

- `implementation-planning` skill
- `implementation_analyzer` (Cortex Analyst) / SQL - Trading costs, market impact, ADV and settlement from SAM_IMPLEMENTATION_VIEW
- `portfolio_analyzer` (Cortex Analyst) / SQL - Current position values and portfolio AUM

Expected Response:

- Trade table for Apple, Microsoft and (usually) NVIDIA: current and target weight, sell notional in dollars, estimated cost in dollars and basis points, execution strategy (VWAP or TWAP) and settlement (T+2)
- Target weights vary between runs - typically just below the breach line or down to the 6.5% warning level - but the shape is stable: **Apple takes the largest trim, Microsoft the smallest, NVIDIA is held or trimmed opportunistically**
- Sizing tied to liquidity: participation capped as a share of ADV, so the agent may phase the trades and flag where a name would take many days to work
- Day-by-day or phased timeline, anchored to the policy remediation window from the breach alert date
- Guidance to redeploy proceeds outside Information Technology, with the effect on the sector weight
- Offers a counterfactual to confirm post-trade weights, or an IC memo PDF

What is not delivered: the plan does not include a tax-lot or capital-gains analysis, tracking-error or risk-budget figures, or a named committee approval workflow. If asked, follow up with the offered counterfactual (`run_counterfactual`) for post-trade weights, or the Counterfactual What-If Analysis scenario for what-if analysis.

Talking Points:

- **Complete Investment Workflow**: From analysis to specific trades with dollar sizes and costs
- **Data-Grounded Execution**: Costs, ADV and settlement come from the implementation data, not from generic assumptions
- **AI-Powered Decision Support**: Research conviction carried through into trade sizing

Key Features Highlighted:

- **End-to-End Investment Process**: Complete workflow from analysis to implementation
- **Specific Action Planning**: Dollar amounts, cost estimates and timelines
- **Liquidity-Aware Sizing**: Trade phasing driven by ADV

Scenario Wrap-up

Business Impact Summary:

- **Time Savings**: Reduced portfolio analysis time from hours to minutes
- **Risk Management**: Proactive identification of concentration issues and active breaches
- **Decision Quality**: Enhanced decision-making through integrated quantitative and qualitative insights
- **Operational Efficiency**: Single interface replacing multiple legacy systems and manual processes

Technical Differentiators:

- **Semantic Understanding**: Natural language queries automatically converted to complex SQL
- **Direct Warehouse Access**: Holdings, alerts and cost data read from governed month-end data with no extracts or reconciliation step
- **AI-Powered Search**: Intelligent document search over broker research and firm policy
- **Multi-modal Analysis**: Seamless combination of structured portfolio data with unstructured research content